SC Honors Geometry - Nappier

4.5 – Dilations • Homework — Odd Answers

Name: _____

Find the image after a dilation centered at the origin with the given scale factor.

1. (3, −2), k = 4

Answer: (12, −8)

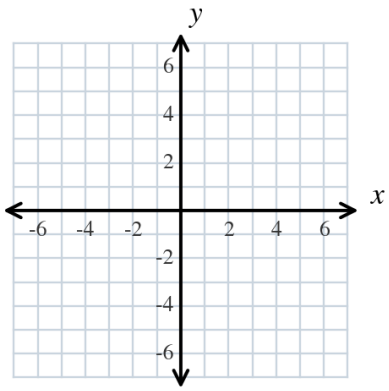
2. (−8, 6), k = 1/2

3. (5, −3), k = −1

Answer: (−5, 3) — the same as a 180° rotation about the origin

Use the coordinate planes provided.

4. Graph △PQR with P(1, 2), Q(3, 5), and R(4, 1) and its image after a dilation centered at the origin with k = 2. List the image coordinates.



5. Graph △STU with S(−4, 2), T(2, 6), and U(6, −2) and its image with k = 1/2. List the image coordinates.

Answer: S′(−2, 1), T′(1, 3), U′(3, −1)

Find the scale factor of each dilation centered at the origin.

6. A(2, 5) → A′(8, 20)

7. B(−6, 9) → B′(−2, 3)

Answer: k = 1/3

Answer each question.

8. Dilate (3, 4) about the center C(−1, 2) with k = 2.

9. A dilation has k = 5. Its preimage has a side of length 3 and an angle of 40°. Give the matching side length and angle in the image.

Answer: side 15, angle still 40° — a dilation scales length but preserves angle measure

10. Is every dilation a congruence transformation? Explain using the word rigid motion.